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AIM : Data Wrangling with pandas: Manipulate and transform data to suit the analysis

Theory :

1 . What is data wrangling? Difference between data wrangling and data preprocessing?

ans.

Data wrangling is the process of cleaning, transforming, organizing and structuring raw data into a suitable format for analysis. It involves handling missing values, correcting inconsistencies, filtering unwanted data and converting data types. Data preprocessing is a broader step mainly used before applying machine learning models and includes tasks such as normalization, encoding categorical values and scaling features. In simple terms, data wrangling prepares raw data for analysis, while data preprocessing prepares cleaned data for modeling and prediction tasks.

2 . Importance of Exploratory Data Analysis (EDA) in Pandas?

ANS:

Exploratory Data Analysis (EDA) is important because it helps understand the structure, quality and patterns present in the dataset before performing detailed analysis or building models. Using Pandas, EDA allows users to inspect datasets, summarize statistics, detect missing values, identify outliers and understand relationships between variables. This process improves decision making and ensures the dataset is suitable for further analysis, which increases the accuracy of results.

3 . Differences between Series, DataFrame and Panel?

ANS:

A Series is a one-dimensional labeled data structure that stores a single column of data with an index. A DataFrame is a two-dimensional data structure that stores data in rows and columns like a table and is the most commonly used structure in Pandas. A Panel is a three-dimensional data structure used to store multiple DataFrames together, but it is now deprecated in modern Pandas versions and replaced by other tools like multi-index

DataFrames. These structures help organize data based on complexity and analysis requirements.

4 . Use of isnull(), notnull(), dropna(), fillna()?

ANS:

The isnull() function is used to detect missing values in a dataset by identifying empty entries. The notnull() function performs the opposite task by identifying values that are present and valid in the dataset. The dropna() function removes rows or columns that contain missing values, which helps clean incomplete data. The fillna() function replaces missing values with suitable alternatives such as mean, median, mode or user-defined values. These functions are essential for maintaining data quality before analysis or modeling.

5 . What is column transformation and why is it important in data analysis?

ANS:

Column transformation is the process of modifying or converting column values into a more useful format for analysis. It includes changing data types, scaling values, encoding categorical data or creating new columns from existing ones. Column transformation is important because raw data is often not in the correct format required for analysis or machine learning. Proper transformation improves data consistency, enhances interpretation and increases the performance of analytical models, making results more reliable and meaningfu

```
import pandas as pd
import numpy as np
data = {
    "Name": ["Amit", "Aditya", "Uday", "krishana", None],
    "Age": [20, 21, None, 19, 22],
    "Marks": [85, 90, 78, None, 88],
    "City": ["Pune", "Mumbai", "Pune", "Russia", "Mumbai"]
}
df = pd.DataFrame(data)
print("Original Data:\n", df)
# Check missing values
print("\nMissing Values:\n", df.isnull())
# Fill missing values
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(0)
df["Name"] = df["Name"].fillna("Unknown")
# Drop rows with any remaining null values
df = df.dropna()
# Change data type
df["Age"] = df["Age"].astype(int)
```

```
# Rename columns
df.rename(columns={"Marks": "Score"}, inplace=True)
# Add new column (Transformation)
df["Result"] = df["Score"].apply(lambda x: "Pass" if x >= 40 else "Fail")
# Filter data
filtered_df = df[df["Score"] > 80]
# Sort data
sorted_df = df.sort_values(by="Score", ascending=False)
# Grouping data
grouped = df.groupby("City")["Score"].mean()
# Display results
print("\nCleaned Data:\n", df)
print("\nFiltered Data (Score > 80):\n", filtered_df)
print("\nSorted Data:\n", sorted_df)
print("\nGrouped Data (Average Score by City):\n", grouped)
```

```
Original Data:
        Name   Age  Marks    City
0      Amit  20.0   85.0    Pune
1    Aditya  21.0   90.0  Mumbai
2      Uday   NaN   78.0    Pune
3  krishana  19.0    NaN  Russia
4      None  22.0   88.0  Mumbai

Missing Values:
     Name    Age  Marks   City
0  False  False  False  False
1  False  False  False  False
2  False   True  False  False
3  False  False   True  False
4   True  False  False  False

Cleaned Data:
        Name  Age  Score    City Result
0      Amit   20   85.0    Pune   Pass
1    Aditya   21   90.0  Mumbai   Pass
2      Uday   20   78.0    Pune   Pass
3  krishana   19    0.0  Russia   Fail
4   Unknown   22   88.0  Mumbai   Pass

Filtered Data (Score > 80):
       Name  Age  Score    City Result
0     Amit   20   85.0    Pune   Pass
1   Aditya   21   90.0  Mumbai   Pass
4  Unknown   22   88.0  Mumbai   Pass

Sorted Data:
        Name  Age  Score    City Result
1    Aditya   21   90.0  Mumbai   Pass
4   Unknown   22   88.0  Mumbai   Pass
0      Amit   20   85.0    Pune   Pass
2      Uday   20   78.0    Pune   Pass
3  krishana   19    0.0  Russia   Fail

Grouped Data (Average Score by City):
```

```
 City
Mumbai    89.0
Pune      81.5
Russia     0.0
Name: Score, dtype: float64
```

Conclusion :

Data wrangling with Pandas is an important step in preparing raw data for analysis by cleaning, organizing and transforming it into a meaningful format. It helps handle missing values, detect inconsistencies and improve the overall quality of the dataset. Exploratory Data Analysis supports better understanding of data structure and patterns before applying further techniques. Structures like Series and DataFrame allow efficient storage and manipulation of data, while functions such as isnull(), notnull(), dropna() and fillna() help manage missing values effectively. Column transformation further improves data usability by converting it into a suitable format for analysis, leading to more accurate and reliable results in data science tasks